

CP2 (Section A) (Section B will be provided on wp)

MCQ:

20x1=20

1. String s= "Hello"; s=s.substring(s.compareToIgnoreCase("hEllo"), s.indexOf("Shinjon".charAt(5))+1); s=?

- a) Hello
- b) hEllo
- c) Hell
- d) Shinjon

2. System.out.println(Math.pow(2, Math.max(Math.round(3.5), Math rint(4.5))));

- a) 16.0
- b) 32.0
- c) 1.0
- d) 2.0

3. String s= "Hello"; int a=10; double b=20.0;

System.out.println(1+2+ '3' + "4" + 5 + (6+7));

- a) 334567
- b) 64513
- c) 544513
- d) 544567

4) for(boolean i=true; i ; i= !i){
 System.out.println(!i ? "HI": "Hello");
}

- a) Hi
- b) Hello
- c) Infinite loop
- d) Error

5) Find order of the array

int array[][]={{2},{4},{5},{6}};

- a) 1x4
- b) 4x4
- c) 4x4
- d) 1x1

6) Which one has right to left associativity

- a) +
- b) -
- c) /
- d) =

7 int day = 5;

```
String result = switch (day) {  
    case 1, 2, 3, 4, 5 -> "Weekday";  
    case 6, 7 -> "Weekend";  
    default -> "Invalid day";  
};
```

System.out.println(result);

- a) Weekday
- b) WeekdayWeekendInvalid day
- c) prints nothing
- d) This is weird

8. `int a[]={5,10,15,20};`

```
    for(int i : a){  
        System.out.print(i+ " ");  
    }
```

- a) 0 1 2 3 b) 5 10 15 20
c) 3 2 1 0 d) Lemme call James Gosling

9. `Integer ob=new Integer(20);` is an example of

- a) Autoboxing b) Unboxing
c) Boxing d) None

10. `Integer ob= 20;`

- a) Autoboxing b) Unboxing
c) Boxing d) None

11. A student wants to store different copies of x and y through a class. What type of members should he use ?

- a) static b) non static
c) private d) let the student decide

12. `class Fruit{`

```
    int a,b;  
    Fruit(int a, int b){  
        this.a=a;  
        this.b=b;  
    }
```

`}`
It's a

- a) default constructor b) Parameterized Constructor
c) Copy Constructor d) Logical Error

13) Non static members are

- a) instance members b) class variables
b) final variables d)

14) `int a=10, c=0;`
`int k=(c=(++a));`
`System.out.println(c);`

- a) 0 b) error c) 10 d) 11

***** Default values are only assigned to class members and arrays.**

```
15) class Debadori{
    void yo(){
        int a;
        System.out.println(a);
    }

    public static void main(String args[]){
        Debadori obj= new Debadori();
        obj.yo()
    }
}
```

- a) error b) 0
- c) Garbage Value d) None

```
16 class Debadori{

    int a;
    void yo(){

        System.out.println(a);
    }

    public static void main(String args[]){
        Debadori obj= new Debadori();
        obj.yo();
    }
}
```

- a) error b) 0
- c) Garbage Value d) None

```
17. class Dorubo {

    void yo(){

        int a[]=new int[3];
        System.out.println(a[0]);
    }

    public static void main(String args[]){

        Dorubo obj= new Dorubo();

        obj.yo();

    }
}
```

- a) error b) 0
- c) Garbage Value d) None

18) String s= "MAGIC";
s.toLowerCase();
System.out.println(s)

- a) MAGIC
- b) magic
- c) Magic
- d) mAGIC

19. Does java compiler passes default constructor if user defined constructor is present

- a) Yes
- b) No
- c) No Idea
- d) 😊

20. Choose the data types that can be used as switch-case labels since Java 7

- a) int
- b) float
- b) String
- d) char

SAQ. Answer the following (Showing dry run is compulsory for snippets)

10x2=20

1. int[] a = {1, 2, 3, 4, 5};
int start = 0;
int end = a.length - 1;
while (start < end) {
int c = a[start];
a[start] = a[end];
a[end] = c;
start++;
end--;
}
print the array (a)

2. Write prototype of a function that takes String as argument and returns a char

3. Write the following expression in JAVA.

$$k = \sqrt[4]{|x - y|}$$

4. int a= 3,b=2;
b*= b * b/b*b + ++a;
b=?
a=?

5. Find output and the number of times the loop executes

```
long n=1234567890, c=0;

while(n>0){

    long d= n/Math.pow(10,c++);

    System.out.println(d);

    n/=10;

}
```

6. find output and the number times the loop executes

```
int p=1;
do{
    System.out.println(p);
    p*=2;

}while(p!=32);
```

7. For reversing a number using digit extraction algorithm, we need to extract digit and store it in d, then use the statement $r=r*10+d$. Can we directly write $r*=10+d$? . Justify your answer.

8. convert to switch case

```
if(a==1 && b==1){

    System.out.println("Hi");

}else if(a==1 && b==2){

    System.out.println("Hello");

}
```

9.

```
class Pakhi{
    static <T> void PrintAnything(T n){
        System.out.println(n);
    }
    public static void main(String args[]){

        PrintAnything(2);
        PrintAnything (1.5);
        PrintAnything("Hi");
    }
}
```

```
10. int i=10;
for( ; ;){
    if(i==i)break;
}
```

How many times will this loop execute ? What if ***if(i==i)break;*** is removed ?